

Conceptual Answers

1. Request Lifecycle

When curl sends a POST request to `/users`, the request first reaches the embedded Tomcat server. Tomcat forwards it to Spring MVC's `DispatcherServlet`, which acts as the front controller. The `DispatcherServlet` consults the `HandlerMapping` to locate the correct controller method and uses a `HandlerAdapter` to invoke that method. The controller then calls the service layer, which calls the repository layer, and the response is returned back through the same chain to the client.

2. Serialization

Spring Boot uses the Jackson library to convert JSON into Java objects. When a JSON request body is received, Jackson maps JSON keys to Java field names. If the JSON key is `upiId`, it maps correctly to the Java field `upiId`. If the key is `upi_id`, mapping fails unless `@JsonProperty("upi_id")` is used.

3. Spring Boot Features

1. Embedded Server – Tomcat runs inside the application without external installation.
 2. Auto Configuration – Spring Boot automatically configures H2, JPA, and Spring MVC.
 3. Production Ready Defaults – Health checks, metrics, logging, and sensible defaults are automatically available.
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4. Spring vs Spring Boot

Using plain Spring would require manual configuration of the servlet container, `DispatcherServlet`, `datasource`, `transaction manager`, `JPA configuration`, and `dependency wiring`. Spring Boot automatically configures these components using `starter dependencies` and `auto-configuration`. This significantly reduces boilerplate code and setup effort.

5. Stateless REST

Stateless means each HTTP request contains all information required to process it, and the server does not store client session state between requests. This allows requests to be handled by any server instance. If PayFlow runs on multiple servers behind a load balancer, statelessness ensures scalability and consistency because no request depends on previous interactions.

6. Persistence

If transactions were stored in a Java List, all records would exist only in memory. Restarting the application would clear the List and permanently lose all transaction data. Databases persist data on disk, ensuring records survive application restarts. For a payments application, losing transaction history would be unacceptable because financial records must be retained reliably.